

A shared item representation forces a three-way trade-off in neural IRT  
the gradient flow predicts it; the experiments confirm it

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## Setup: amortized neural IRT

- Encoder (LSTM): response history  $\rightarrow$  ability  $\theta_t$
- Item embedding  $\rightarrow$  discrimination  $\alpha_i$ , difficulty (step thresholds)  $\beta_i$ ; GPCM decoder

$$P(Y_i = k \mid \theta) = \frac{\exp \sum_{m=1}^k \alpha_i (\theta - \beta_{i,m})}{\sum_{l=0}^{K-1} \exp \sum_{m=1}^l \alpha_i (\theta - \beta_{i,m})} \quad (2\text{PL at } K=2)$$

- Trained on response prediction only. No supervision on  $\theta, \alpha, \beta$ .
- Scale is gauge-free  $\Rightarrow$  recovery = rank (Spearman  $\rho$ ).

## The item code has two consumers

- a **value** embedding  $v_q$  (narrow)  $\rightarrow$  encoder input  $\rightarrow$  **ability**  $\theta$
  - a **key** embedding  $k_q$  (wide)  $\rightarrow$  **discrimination**  $\alpha$ , **difficulty**  $\beta$  readouts
  - **Shared**: one embedding, width  $W$ , feeds both consumers
  - **Decoupled**: a narrow value embedding + a separate wide key embedding
- $\Rightarrow$  *shared ties the encoder-input width to the readout width.* That tie is the problem.

## First the math: $\alpha$ is the slow eigen-mode (derived)

Gradients ( $r = p - y$ ):  $\partial_\theta L = r\alpha$ ,  $\partial_\beta L = -r\alpha$ ,  $\partial_\alpha L = r(\theta - \beta)$ .

The per-response score is one vector  $s = (\alpha, \theta - \beta, -\alpha)$ , so the Fisher  $F = w s s^\top$  ( $w = p(1-p)$ ) is **rank one**. The  $(\theta, \alpha)$  block has exact eigenvalues

$$\lambda_{\pm} = \frac{1}{2} [(l_\theta + l_\alpha) \pm \sqrt{(l_\theta - l_\alpha)^2 + 4l_{\theta\alpha}^2}], \quad \lambda_- \approx l_\alpha(1 - \rho^2).$$

- the slow mode is  $\alpha$ :  $l_\alpha = \mathbb{E}[w(\theta - \beta)^2]$  is suppressed ( $w$  peaks where  $\theta = \beta$ )
- coupling  $\rho^2 = l_{\theta\alpha}^2 / (l_\theta l_\alpha)$  worsens it:  $\kappa_2 = l_\theta / [l_\alpha(1 - \rho^2)] \geq l_\theta / l_\alpha$

The math singles out  $\alpha$  before any experiment.

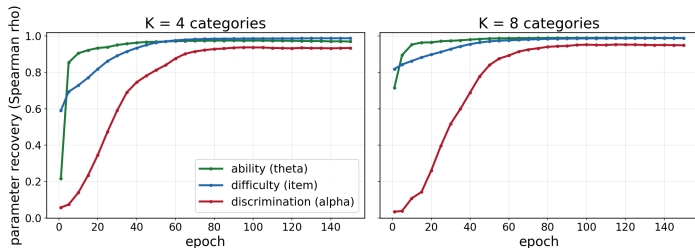
## What the math predicts (then we test each)

- ① **Discrimination recovers last** (it is the single slow mode;  $\theta, \beta$  fast).
- ② **A shared code forces a three-way trade-off**:  $\theta$  wants it narrow,  $\alpha$  wants it wide,  $\beta$  is indifferent.
- ③ **Sharing adds conditioning**  $\kappa = I(\theta)/I(\alpha)$ : it throttles  $\alpha$ 's rate; decoupling removes it (speedup  $\sim \kappa$ , growing with  $K$ ).
- ④ **Positivity is necessary, the map shape is not**: among smooth positive maps the convergence is the same; exp is not special.

Synthetic GPCM, known  $\theta, \alpha, \beta$ , LSTM. Each prediction is tested next.

# Prediction 1: discrimination recovers last

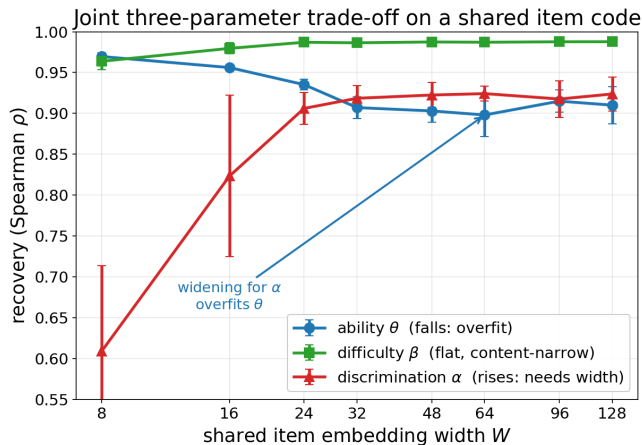
Ability and difficulty are learned fast; discrimination lags



- ability, difficulty: fast (high Fisher rate)
- discrimination: slow, lags by hundreds of epochs

Exactly the ordering the rates predict.

## Prediction 2: the three-way trade-off



As the shared code widens:

- $\alpha$  rises 0.61  $\rightarrow$  0.92 (gets capacity)
- $\theta$  falls (overfits the wide input)
- $\beta$  flat-high 0.96  $\rightarrow$  0.99

One knob, opposed pressures.  $\beta$  does not pay.

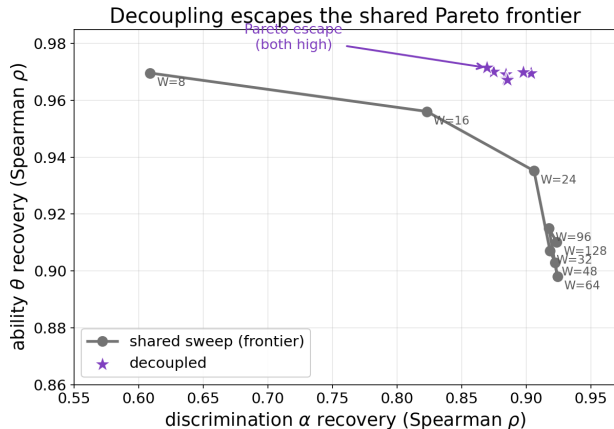
## The two pressures are of different KIND

- $\alpha$ : **a RATE effect, endpoint invariant.**  $\alpha$  is a free per-item table, so the optimum is unbiased (free-table invariant). Sharing only slows it (conditioning  $\kappa$ ); decoupling speeds it. Transient: vanishes as training  $\rightarrow \infty$ , not at a fixed budget. (P9)
- $\theta$ : **an ENDPOINT effect, not invariant.**  $\theta$  is read by the amortized encoder, NOT a free table. A wide input overfits it,  $\text{Var}(\hat{\theta}) \sim \sigma^2 W/n$ . It does NOT vanish at convergence. (P10)
- $\beta$ : **indifferent control.** High Fisher, non-pooled  $\Rightarrow$  neither needs width nor is hurt ( $\Delta_\beta \approx 0$ ). (P11)

A slow mode (rate) *and* an overfit (endpoint), tied by one width.



## Prediction 3a: not capacity, allocation

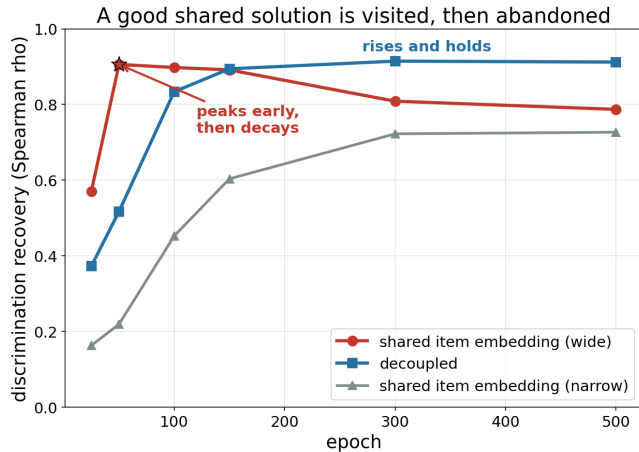


**Control:** matched total item-embedding params.

- widening the shared code traces a Pareto frontier
- decoupled sits **above** it, even at  $6\times$  the shared budget

$\Rightarrow$  allocation, not budget. Decoupling escapes the frontier (P12).

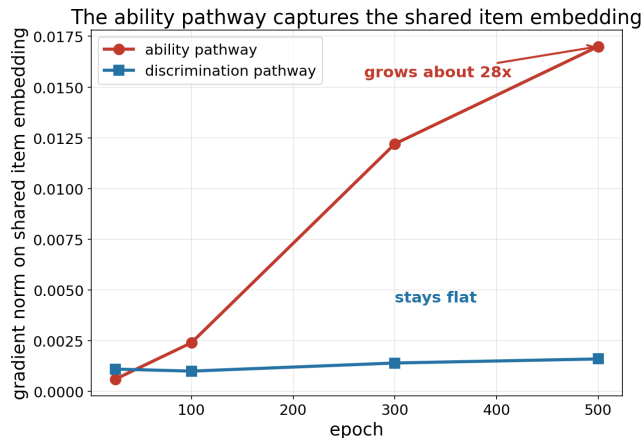
## Prediction 3b: the good $\alpha$ solution is reached then left



- wide-shared  $\alpha$ : peaks 0.91 (ep 50), decays to 0.79 (ep 500)
- decoupled: rises 0.91, holds

$\Rightarrow$  reachable, reached, then abandoned  
= dynamics, not a ceiling.

## Prediction 3c: the shared code is conditioned



- $\theta$ -pathway pull grows  $\sim 28\times$ ;  
 $\alpha$ -pathway flat
- $\cos(g_\theta, g_\alpha) \approx 0$  (not a tug-of-war)
- the lag is eigenvalue spread,  
stiffness  $\kappa \approx 1.9$  at  $K=4$

## Prediction 3: the gradient-flow ODE

One item's embedding  $e_q$ . Near its optimum,  $\delta = e_q - e_q^*$ ,  $H_q$  the Hessian in  $e_q$ 's directions:

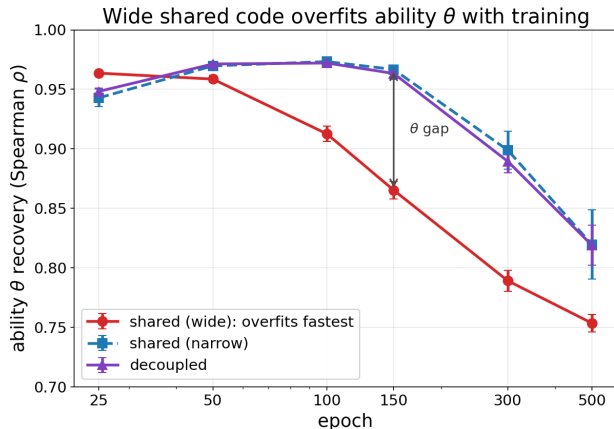
$$\dot{\delta} = -H_q \delta, \quad \delta_c(t) = \delta_c(0) e^{-\lambda_c t}.$$

- Shared:  $e_q$  feeds encoder ( $\theta$ ) and  $\alpha, \beta$  readouts  $\Rightarrow$  steep  $\theta$  direction ( $\sim I(\theta)$ ), flat  $\alpha$  direction ( $\sim I(\alpha)$ ):  $\kappa = I(\theta)/I(\alpha) \gg 1$
- GD,  $\delta_c(t) = (1 - \eta\lambda_c)^t \delta_c(0)$ , stable  $\eta = 1/\lambda_{\max}$ : slow mode  $(1 - 1/\kappa)^t \Rightarrow t_{\alpha}^{\text{sh}} = \kappa \log(1/\epsilon)$ ; decoupled  $O(\log \frac{1}{\epsilon})$

$\text{speedup} = O(\kappa), \quad \kappa = I(\theta)/I(\alpha) \text{ (grows with } K\text{)}$

Same fixed point  $\Rightarrow$  rate, not endpoint. Largest under GD; Adam compresses  $\kappa$ .

## The $\theta$ side: an endpoint overfit



- wide-shared  $\theta$  overfits worst:  $0.96 \rightarrow 0.75$
- decoupling keeps the encoder narrow  $\Rightarrow$  protects  $\theta$  at the operating point

Bias and variance:  $\text{Var}(\hat{\theta}) \sim \sigma^2 W/n$ . An **endpoint** loss (not a rate), outside the free-table invariant. (The decay with long training, even narrow, is a separate regularization issue, not the width effect.)

## The resolution: untie the two widths

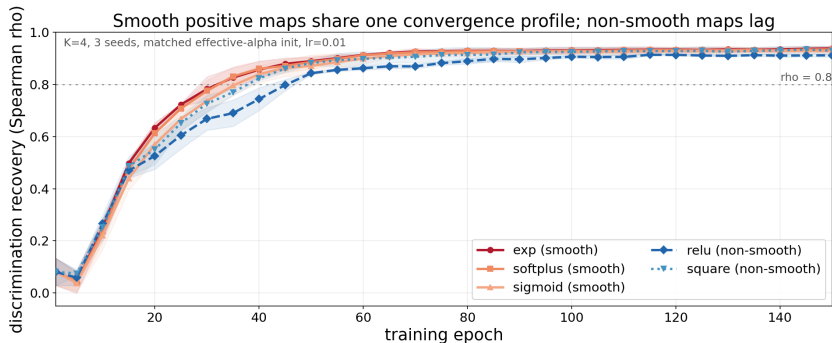
- **Shared:** one width, opposed knobs ( $d\alpha/dW > 0$ ,  $d\theta/dW < 0$ ,  $d\beta/dW \approx 0$ )  $\Rightarrow$  a Pareto frontier
- **Decoupled:** two independent widths  $\Rightarrow$  narrow encoder (good  $\theta, \beta$ ) + wide readout (good  $\alpha$ )

$\Rightarrow$  decoupling is Pareto-**dominant**: it escapes the frontier, not moves along it. (P12)

It cures both pressures at once:  $\theta$  keeps its bottleneck,  $\alpha$  gets its capacity.

## Prediction 4: positivity needed, map shape is not

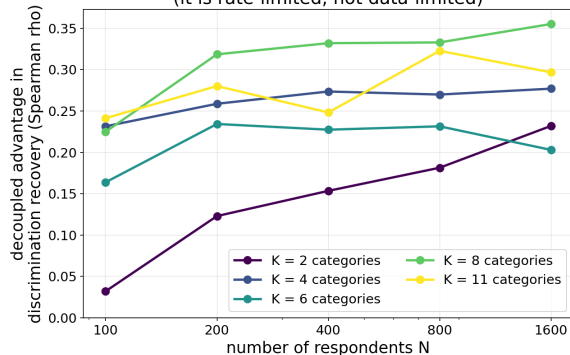
**Derived (Thm P7):** for smooth monotone  $g$ ,  $m_g(\alpha)=[g'(g^{-1}(\alpha))]^2 > 0$  is a positive time-change of one canonical flow  $\Rightarrow$  same fixed point and rank. exp is not special.



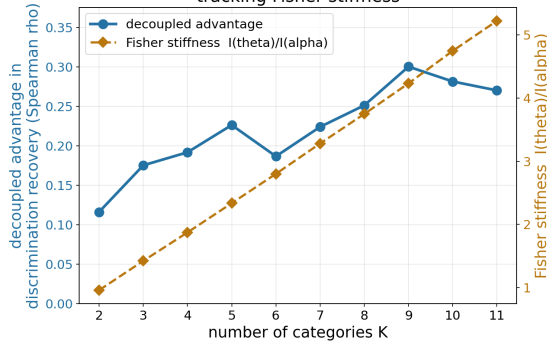
- confirmed: smooth maps share one profile; non-smooth lag (relu dead zone  $m_g=0$ ; square non-injective)
- direct- $\alpha$  control ( $\theta, \beta$  frozen): a scalar reparameterization changes only speed, not the fixed point ( $\kappa=1$ )

# The $\alpha$ rate effect: finite-sample and $K$ -scaled

More data does not shrink the advantage  
(it is rate-limited, not data-limited)



The advantage grows with the number of categories  $K$ ,  
tracking Fisher stiffness



- Fixed budget, vary  $N$ : the decoupling advantage does NOT shrink with data (rate-limited).
- Across  $K=2$  to 11: it tracks the Fisher stiffness  $\kappa$ ,  $\rho = 0.89$ .



# What we proved (results and insights)

- **The map is a preconditioner, not a lever.** Smooth monotone positive maps share one canonical flow, so the fixed point and rank are map-invariant (Thm P7). Positivity and smoothness matter; exp does not.
- **$\alpha$  is the rank-one Fisher's slow direction.**  $I_\alpha = \mathbb{E}[w(\theta - \beta)^2]$  is suppressed where responses concentrate; coupling makes  $\kappa_2 = I_\theta/[I_\alpha(1 - \rho^2)] \geq I_\theta/I_\alpha$ .
- **Sharing is one ill-conditioned code block.** A single step size resolves  $\alpha$  in  $t_\alpha = \kappa \log(1/\epsilon)$  iterations; decoupling gives  $O(\log \frac{1}{\epsilon})$ . Speedup  $O(\kappa)$ , growing with  $K$ .
- **Two effects of different kind.**  $\alpha$  a convergence-RATE effect (endpoint-invariant);  $\theta$  an ENDPOINT overfit ( $\text{Var} \sim W/n$ );  $\beta$  indifferent.

# Conclusion

The gradient flow predicts, and the experiments confirm, that a shared item code forces a three-way trade-off:  $\theta$  overfits wide,  $\alpha$  needs wide,  $\beta$  is content-narrow. Decoupling unties the widths and dominates.

- Two mechanisms of different kind:  $\alpha$  a convergence-RATE effect (endpoint invariant,  $O(\kappa)$ , grows with  $K$ );  $\theta$  an ENDPOINT overfit ( $\text{Var} \sim W/n$ ).  $\beta$  indifferent.
- A measurement-validity condition: recovered parameters are otherwise representation-allocation artifacts.
- Positivity map ruled out as the lever.

Scope: synthetic GPCM, LSTM (architecture-independent on swap), 3 to 10 seeds.  $\theta$ -overfit formalized for a linear amortizer (LSTM lift argued). Open: NRM slopes, GRM, real data.